Case Study Report

Clustering Analysis on East-West Airlines Customers

K-Means,Hierarchical and DBSCAN

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1. Problem Statement

East-West Airlines wants toknow more about its frequent flyer customers (how they fly, how they earn miles, how they use the credit cards). I'll try to find customer groups (segments) using 3 clustering algorithms and then compare them.

The data has 3999 customers and 12 columns (shape printed in Task 1 of the notebook), taken from the East-West Airlines workbook. The Description sheet of that workbook says that the key issues are the customers' flying patterns, how they earn and use frequent flyer rewards, and their use of the airline credit card, and that the task is to identify customer segments via clustering. The three algorithms I used are K-Means, Hierarchical clustering and DBSCAN.

My plan was:

1. Load the data and clean it (missing values, outliers, scaling)
2. EDA with a bunch of plots
3. K-Means (elbow curve + silhouette)
4. Hierarchical clustering (different linkages + dendrogram)
5. DBSCAN (trying different eps and min_samples)
6. Compare everythingand write what I understood from the clusters

2. Methodology

This sectionfollowsthesteps I did in the notebook, in order. The task numbers are the section numbers of the notebook.

Task 0 - Importing libraries

Before starting I imported the libraries I needed: pandas and numpy for the data, matplotlib and seaborn for the plots, scikit-learn for scaling, PCA, the three clustering algorithms, the silhouette score and the nearest neighbour search (used for the k-distance plot), and scipy for the linkage matrix and the dendrogram. I also fixed the random seed (42) so that the results can be repeated.

Task 1 - Loading the data

The excel file has 2 sheets - Description (just explains the columns) and data (the actual data). So I read the data sheet.

The data sheet has 3999 rows and 12 columns. The info output of the notebook shows that all 12 columns are integers (int64) and none of them has null values. The first five rows are shown below.

Table 1: First five rows of the data sheet (output of the notebook)

Row	ID#	Balance	Qual_ miles	cc1_ miles	cc2_ miles	cc3_ miles	Bonus_ miles	Bonus_ trans	Flight_ miles_ 12mo	Flight_ trans_ 12	Days_ since_ enroll	Award?
0	1	28143	0	1	1	1	174	1	0	0	7000	0
1	2	19244	0	1	1	1	215	2	0	0	6968	0
2	3	41354	0	1	1	1	4123	4	0	0	7034	0
3	4	14776	0	1	1	1	500	1	0	0	6952	0
4	5	97752	0	4	1	1	43300	26	2077	4	6935	1

The meaning of each column (from the Description sheet) is below.

Column	Meaning
ID#	unique customer id (not useful for clustering)
Balance	miles eligible for award travel
Qual_miles	miles counted for Topflight status
cc1_miles, cc2_miles, cc3_miles	miles earned with the 3 credit cards in last 12 months, but given as **bins** 1-5 (1 = under 5,000 ... 5 = over 50,000)
Bonus_miles	miles from non-flight bonus transactions
Bonus_trans	number of non-flight bonus transactions
Flight_miles_12mo	flight miles in last 12 months
Flight_trans_12	number of flight transactions in last 12 months
Days_since_enroll	days since the customer enrolled
Award?	1 if the customer got an award, else 0

Note to self: cc1/cc2/cc3 are not real miles, they are categories 1 to 5, so I should remember that when reading the plots.

Task 2.1 - Missing values and duplicates

Here I checked every column for missing values and checked the data for duplicate rows and duplicate IDs.

Table 2: Missing values and duplicate check (output of the notebook)

Item	Value
Missing values in any of the 12 columns	0
Duplicate rows	0
Duplicate IDs	0

No missing values and no duplicates, so nothing to impute or drop here. The ID# column is just an identifier, so I'm dropping it because it has no meaning for clustering (it would only add noise).

Task 2.2 - Outliers

I made boxplots of the 7 numeric columns before doing anything, to see how bad the outliers are.

Figure 1: Boxplots of the numeric features before removing outliers

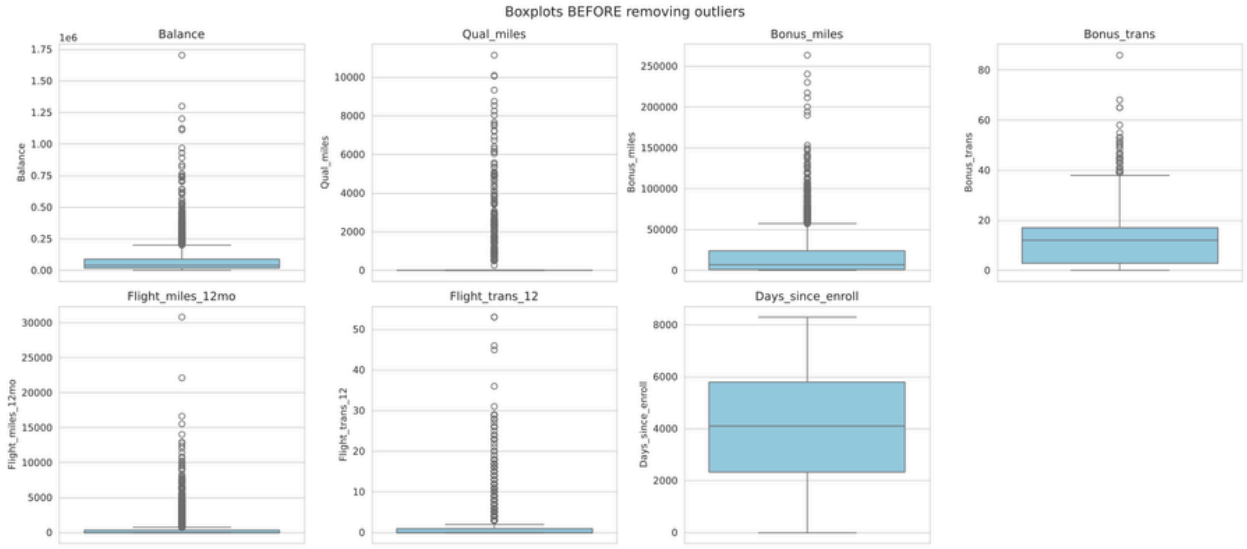


Figure 1. One boxplot for each of Balance, Qual_miles, Bonus_miles, Bonus_trans, Flight_miles_12mo, Flight_trans_12 and Days_since_enroll (y-axis: value of the column).

Interpretation

Lots of points way above the whiskers. This means the columns have a lot of extreme values, so I could not just use the data like this.

So I'm using the **z-score method** instead: remove a row if any numeric column is more than 4 standard deviations from the mean. Let me check how many rows go away for different cutoffs first.

Table 3: Rows that would be removed for different z-score cutoffs (output of the notebook)

Cutoff (z-score)	Rows that would be removed
z > 3	321
z > 4	184
z > 5	125

I went with 4, because it removes less than 5% of the rows.

Table 4: Result of the outlier removal (output of the notebook)

Item	Value
Rows before	3999
Rows after	3815
Rows removed	184 (4.6%)

Task 2.3 - Feature scaling

The columns are on totally different scales (Balance is in the 10,000s, cc2_miles is 1 to 3). K-Means, hierarchical and DBSCAN all use distances, so the big columns would take over everything. I'm using StandardScaler (mean 0, std 1) to fix this.

After scaling I checked the means and standard deviations of all 11 columns. The notebook printed means of about 0 and standard deviations of 1 for every column, which is what StandardScaler should give, so the scaling worked.

Tasks 3 to 7 - how I did the rest

Task 3 (EDA): I looked at the shape of every column with histograms and count plots, at the relationships between columns with a correlation heatmap, scatter plots and a pairplot, and I used PCA to get a 2D picture of the data. The results are in Section 3 of this report.

I can't plot 10 dimensions, so I use PCA to squeeze the scaled data into 2 components. **Important:** the clustering is done on the full scaled data, PCA is just so I can draw the clusters on a 2D plot.

Task 4 (K-Means): I ran K-Means for K from 2 to 10 and used the elbow curve and the silhouette score to choose K, then looked at the average values of each cluster. Task 5 (Hierarchical): I drew a dendrogram with Ward linkage, compared the Ward, complete, average and single linkages for 3 to 6 clusters, and then profiled the Ward clusters. Task 6 (DBSCAN): I used a k-distance plot to get an idea of eps, tried a grid of eps and min_samples values, and then profiled the clusters of the setting I chose. Task 7: I compared the three algorithms with their silhouette scores and plots. Sections 4 and 5 of this report have the results.

3. Statistical Findings

Task 2.1 (continued) - Descriptive statistics

Thisisthedescribetableofthe11columns(withoutID#), before outlier removal.

Table 5: Descriptive statistics of the columns before outlier removal (output of the notebook, shown to 2 decimals)

Column	count	mean	std	min	25%	50%	75%	max
Balance	3999.0	73601.33	100775.66	0.0	18527.5	43097.0	92404.0	1704838.0
Qual_miles	3999.0	144.11	773.66	0.0	0.0	0.0	0.0	11148.0
cc1_miles	3999.0	2.06	1.38	1.0	1.0	1.0	3.0	5.0
cc2_miles	3999.0	1.01	0.15	1.0	1.0	1.0	1.0	3.0
cc3_miles	3999.0	1.01	0.20	1.0	1.0	1.0	1.0	5.0
Bonus_miles	3999.0	17144.85	24150.97	0.0	1250.0	7171.0	23800.5	263685.0
Bonus_trans	3999.0	11.60	9.60	0.0	3.0	12.0	17.0	86.0
Flight_miles_12mo	3999.0	460.06	1400.21	0.0	0.0	0.0	311.0	30817.0
Flight_trans_12	3999.0	1.37	3.79	0.0	0.0	0.0	1.0	53.0
Days_since_enroll	3999.0	4118.56	2065.13	2.0	2330.0	4096.0	5790.5	8296.0
Award?	3999.0	0.37	0.48	0.0	0.0	0.0	1.0	1.0

Looking at the describe table, the columns are super skewed. For example Balance has a mean of ~73k but max of 1.7 million, and Qual_miles is 0 for more than 75% of customers. So outliers are going to be a problem. Let me look at boxplots first.

Task 3.1 - Distribution of the features

Figure2:Histogramsofthenumericfeaturesafter removing outliers

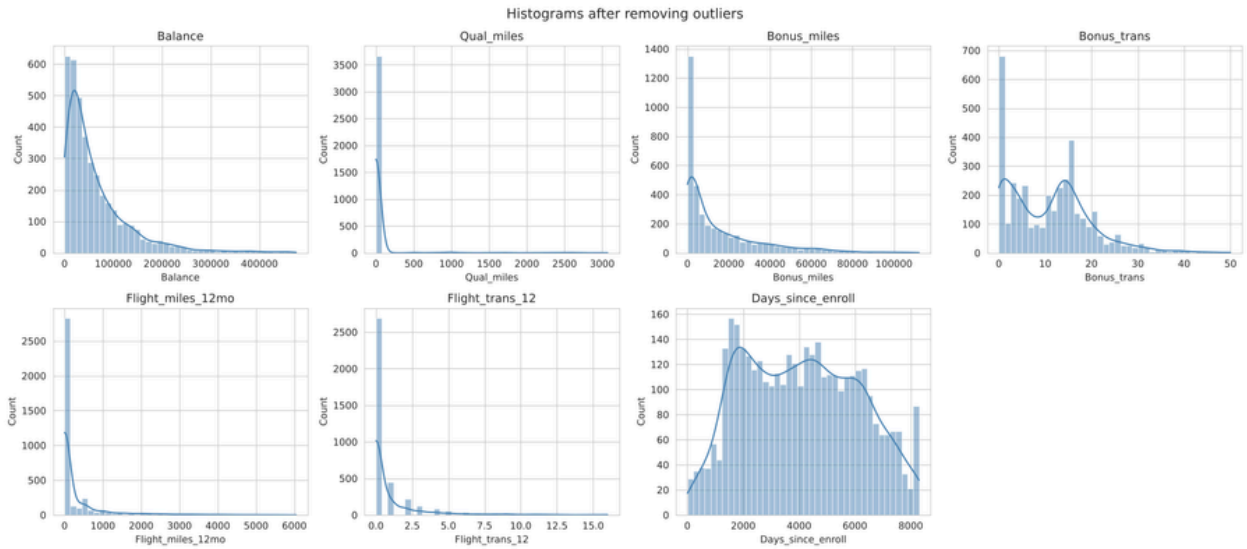


Figure 2. Histogram with a density curve for each numeric column (x-axis: value of the column, y-axis: number of customers).

Interpretation

- Almost all the columns are still right skewed even after removing outliers (skewness of `Qual_miles is about 6, Flight_miles_12mo and Flight_trans_12 about 3.5, Balance` about 2.3).

•Qual_miles is 0 for 96% of the customers and Flight_miles_12mo is 0 for around 71%. So most members hardly fly with the airline at all, which is interesting.

- Bonus_trans looks the closest to a normal shape and Days_since_enroll is spread out almost evenly, so there are both old and new members.
- So the data is mostly a big group of low-activity customers plus a long tail of very active ones.

Task 3.2 - Categorical-type columns (credit card bins and Award)

Figure3:Countsofthecreditcardbinsand ofAward

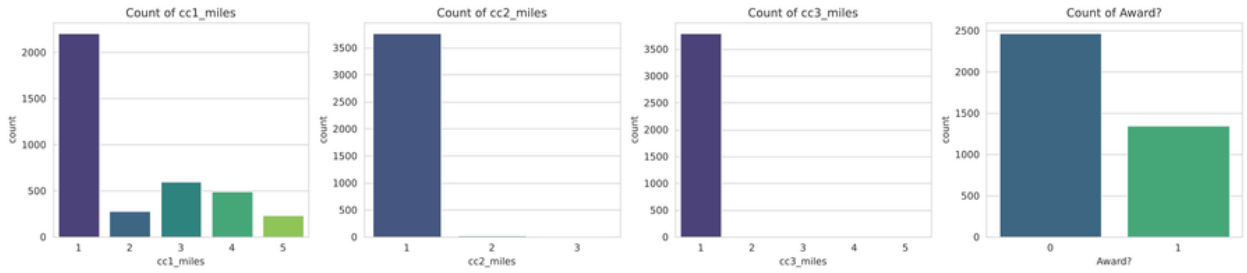


Figure 3. Bar charts of how many customers fall in each value of cc1_miles, cc2_miles, cc3_miles and Award? (x-axis: value, y-axis: count of customers).

Interpretation

•cc2_miles and cc3_miles are 1 for around 99% of the customers, so nearly nobody earns miles with the Rewards or Small Business card. These two columns won't really help in separating groups.

•cc1_miles has more variety, but still about 58% are in bin 1 (under 5,000 miles), and a smaller group is in the higher bins.

•Around 35% of the customers have got an award, 65% have not.

The notebook also printed the award rate of the cleaned data as 0.352, which is the 35% I mentioned above.

Task 3.3 - Correlation heatmap

Figure 4: Correlation heatmap of the features

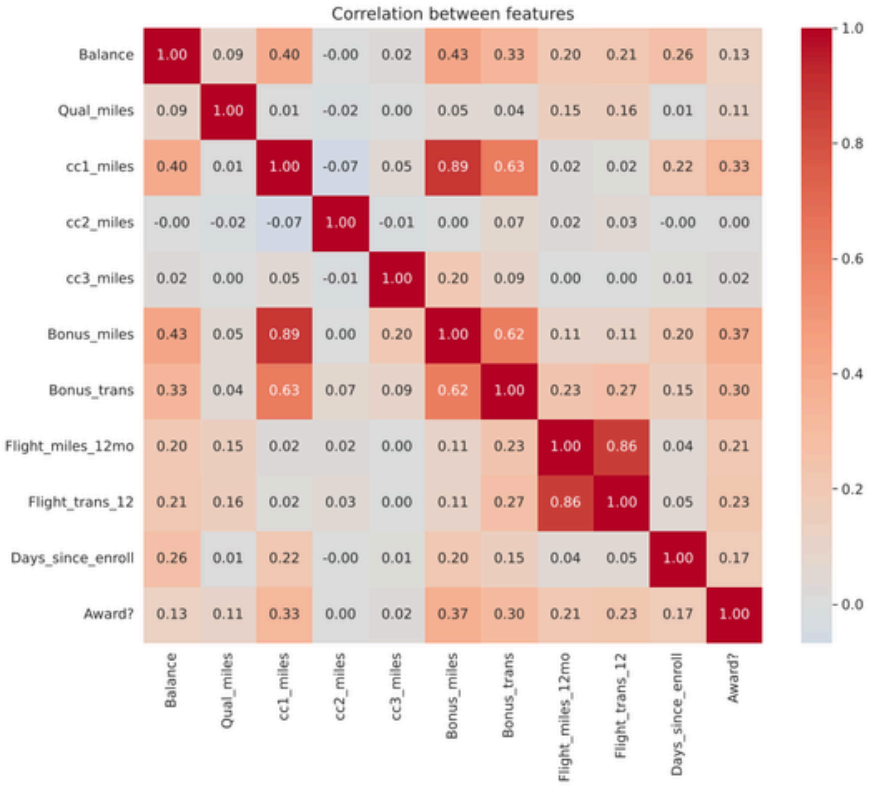


Figure 4. Correlation coefficient between every pair of columns of the cleaned data (colour scale from -0.07 to 1.00).

Interpretation

- Flight_miles_12mo and Flight_trans_12 are strongly correlated (0.86), which makes sense, more flights = more flight miles.
- cc1_miles and Bonus_miles are also strongly correlated (0.89), I think because miles earned using the credit card also show up as bonus miles. Bonus_miles and Bonus_trans have 0.62.
- Balance is related more to Bonus_miles (0.43) and cc1_miles (0.40) than to flight miles (0.20). So customers mostly collect their miles through credit card / bonus transactions, not by flying.
- Qual_miles has weak correlation with everything.
- No pair is so correlated (like 0.95+) that I need to drop a column, so I'm keeping all of them.

Task 3.4 - Looking for possible clusters with scatter plots

Figure 5: Scatter plots coloured by Award

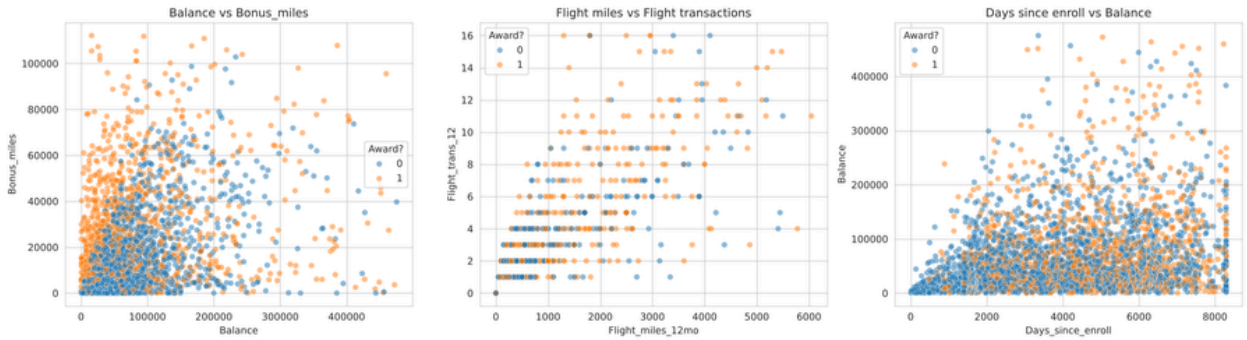


Figure 5. Balance vs Bonus_miles, Flight miles vs Flight transactions, and Days since enroll vs Balance; blue points are customers with Award? = 0 and orange points have Award? = 1.

Interpretation

- There is no clear separate blobs. It is mostly one dense group near the lower left (low balance, low bonus miles, no flights) with a long tail going out.
- In the flight miles vs flight transactions plot, a huge number of points are stuck at (0, 0), and the rest are on a roughly straight line.
- Customers with Award? = 1 (orange) are more towards the higher balance / bonus miles side, but they are still mixed with the others.
- Because of this I don't expect super well separated clusters, so the silhouette scores will probably be low-ish.

Task 3.5 - Pairplot of the main features

Iusedarandomsampleof800rowsforthisplot,because the pairplot is slow with all the rows.

Figure 6: Pairplot of the main features coloured by Award

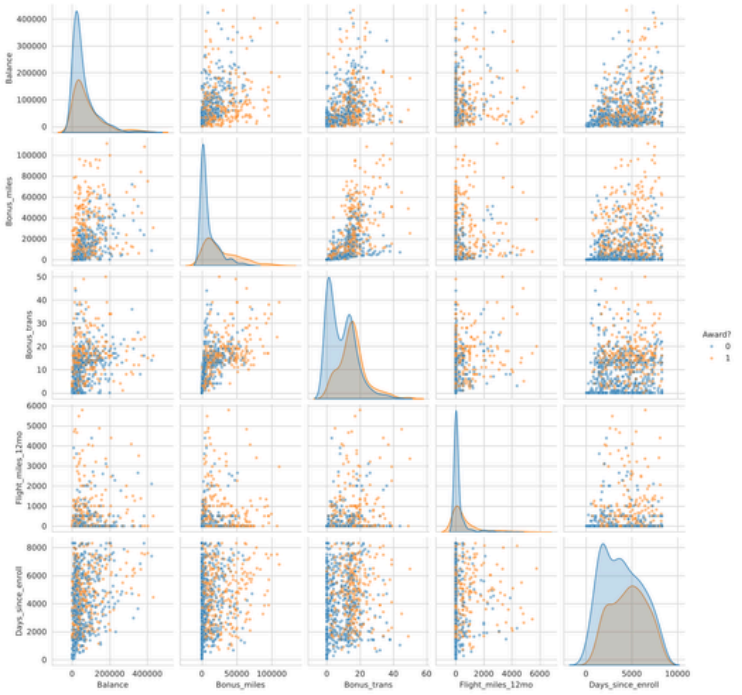


Figure 6. Scatter plots of every pair of Balance, Bonus_miles, Bonus_trans, Flight_miles_12mo and Days_since_enroll, with density curves on the diagonal; blue = Award? 0, orange = Award? 1.

Interpretation

Same story as before. The diagonal (kde) plots are all skewed with a peak near zero, and the scatter plots don't show clean islands of points. The Award? = 1 customers generally have higher Bonus_miles and Bonus_trans, so the Award column might have an effect on the clustering later (I'll keep an eye on it in DBSCAN).

Task 3.6 - Reducing to 2D with PCA

Table 6: Variance explained by the first two principal components (output of the notebook)

Item	Value
PC1	0.289
PC2	0.167
Total (PC1 + PC2)	0.455

Figure 7: The data in 2D (PCA) before clustering

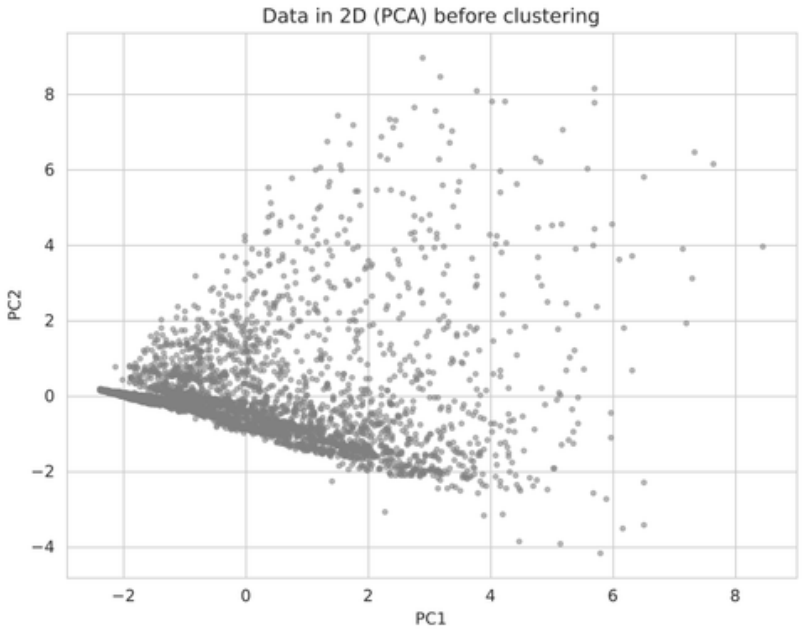


Figure 7. Every customer of the cleaned data plotted on the first two principal components (x-axis: PC1, y-axis: PC2).

Interpretation

PC1 and PC2 together explain only about 45% of the variance, so this 2D picture is losing a lot of information and clusters that look overlapped here may be better separated in the real 10 dimensions. Still, it looks like one big cloud shaped like a wedge, with a sparse tail of very active customers. No obvious gaps, so the clusters will probably touch each other.

4. Model Evaluation

There is no supervised model in this notebook, so the models I evaluate are the three clustering algorithms. To evaluate them I used the silhouette score. It goes from -1 to 1, and a higher value means that the customers are closer to their own cluster than to the other clusters. Values near 0 mean the clusters overlap.

Task 4 - K-Means clustering

Task 4.1 - Choosing K (elbow curve and silhouette score)

Figure8:Elbowcurveandsilhouettescorefor K=2to10

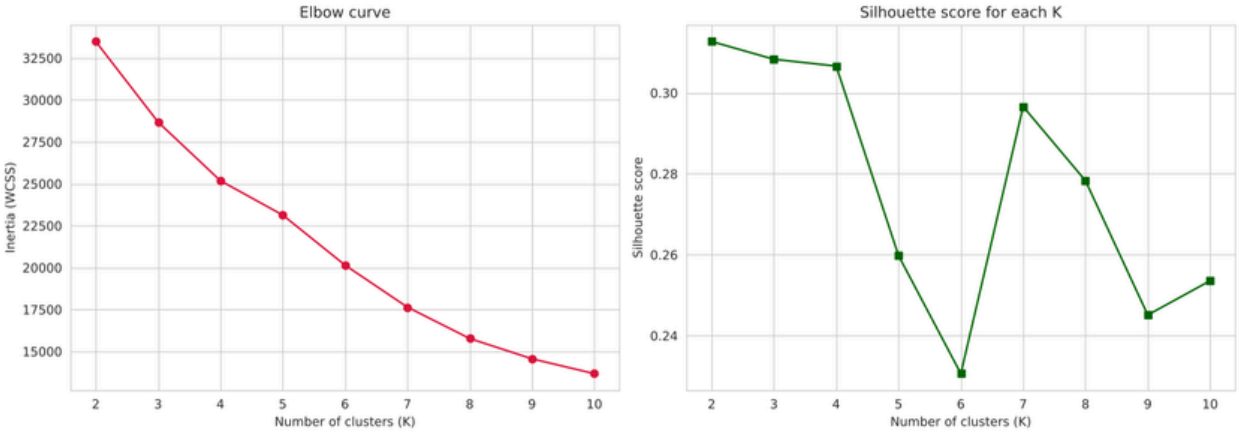


Figure 8. Left: inertia (WCSS) for each number of clusters K. Right: silhouette score for each K.

Interpretation

How I chose K:

- The elbow curve goes down smoothly and there isn't a really sharp elbow. There is a small bend around K = 3 to 4.
- The silhouette score is highest at K = 2 (0.313) but K = 3 (0.308) and K = 4 (0.307) are almost the same. After K = 4 it drops a lot (K = 5, 6 are bad), and there is a small bump at K = 7 but that doesn't help much.
- K = 2 is too simple (just "active" vs "not active"). When I tried K = 4 one cluster had only 12 customers, which is not a useful segment.
- So I'm going with **K = 3**, it is a good balance between the elbow, the silhouette score and being easy to explain.

Table 7: Inertia and silhouette score for each K (output of the notebook)

K	Inertia	Silhouette
2	33486.0	0.313
3	28681.8	0.308
4	25183.0	0.307
5	23135.2	0.260
6	20148.6	0.231
7	17644.1	0.297
8	15773.3	0.278
9	14560.4	0.245
10	13676.4	0.254

I then fitted K-Means with K = 3. The notebook printed a silhouette score of 0.3083 and these cluster sizes:

Table 8: K-Means (K = 3) cluster sizes (output of the notebook)

Cluster	Number of customers
0	2363
1	262
2	1190

Task 4.2 - K-Means cluster visualization

Figure 9: K-Means clusters (K = 3) on the PCA plot

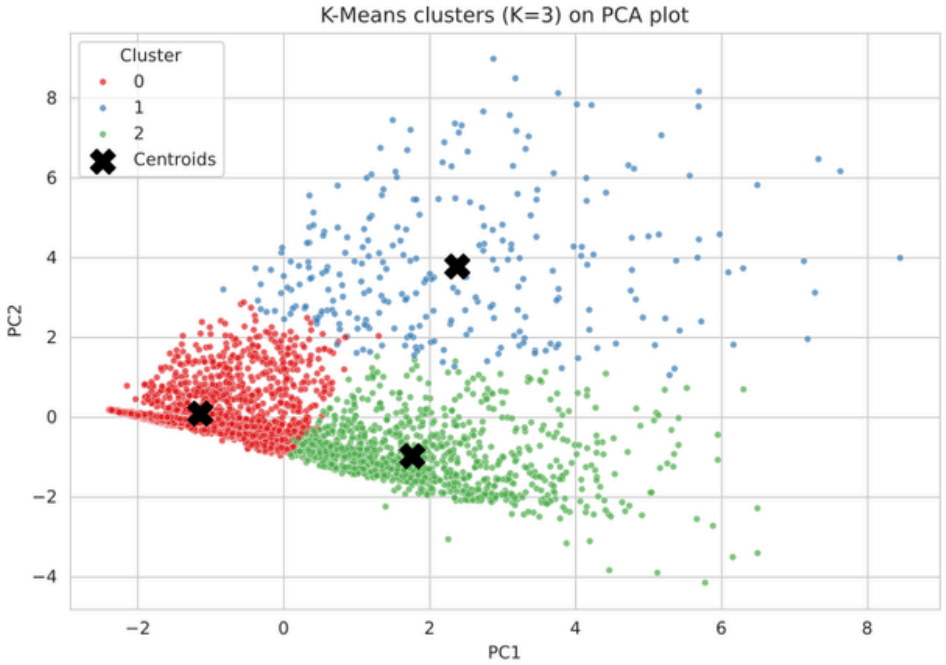


Figure 9. Customers on the first two principal components (x-axis: PC1, y-axis: PC2), coloured by K-Means cluster; the black X marks are the centroids.

Interpretation

In this plot the 3 K-Means clusters sit next to each other and touch at the borders, they are not separate islands. Cluster 0 (red) is the big group at the lower left, cluster 2 (green) is in the middle going to the right, and cluster 1 (blue) is the sparse group in the upper part. This matches the silhouette score of 0.308, which means the clusters are there but they overlap. Also, PC1 and PC2 only explain about 45% of the variance, so some of the overlap may just be because of the 2D view.

Task 4.3 - Understanding the K-Means clusters

Table9:AveragevalueofeachfeatureineachK-Meanscluster, in the original units (output of the notebook)

Cluster	Balance	Qual_ miles	cc1_ miles	cc2_ miles	cc3_ miles	Bonus_ miles	Bonus_ trans	Flight_ miles_ 12mo	Flight_ trans_ 12	Days_ since_ enroll	Award?	Size
0	40124.0	43.6	1.2	1.0	1.0	4247.3	6.6	106.4	0.4	3688.5	0.2	2363
1	107150.3	282.1	1.9	1.0	1.0	19819.5	17.7	2610.4	7.5	4276.4	0.7	262
2	103975.8	43.9	3.7	1.0	1.0	36780.0	18.1	183.8	0.6	4821.0	0.6	1190

Size = number of customers in the cluster. The averages were rounded to 1 decimal in the notebook.

Figure 10: K-Means: average scaled value of each feature per cluster

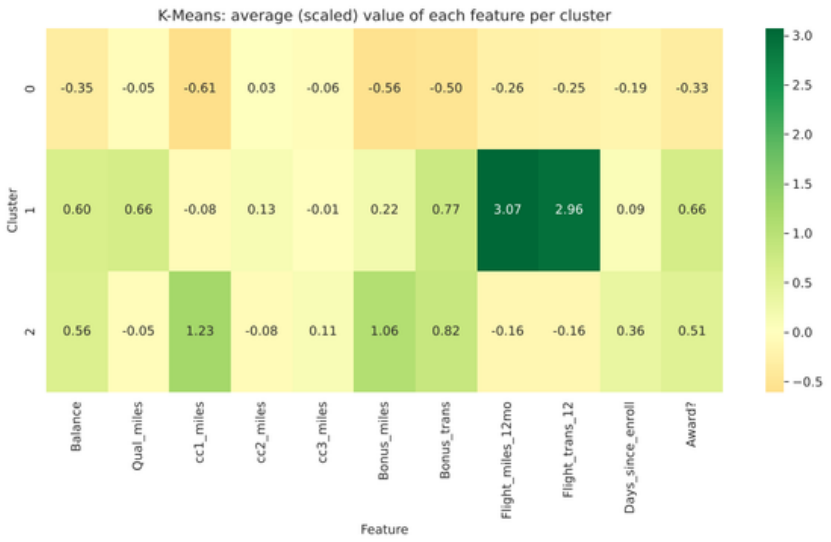


Figure 10. Heatmap of the cluster averages of the scaled features (rows: clusters, columns: features; green = above average, orange = below average).

Interpretation

In this heatmap cluster 1 stands out with very high flight miles and flight transactions (3.07 and 2.96 in scaled units). Cluster 2 has the highest credit card bin (1.23) and bonus miles (1.06) but its flight values are a bit below average (-0.16). Cluster 0 is below average on almost everything (for example balance -0.35 and bonus miles -0.56).

Figure 11: K-Means: Balance, Bonus_miles and Days_since_enroll by cluster

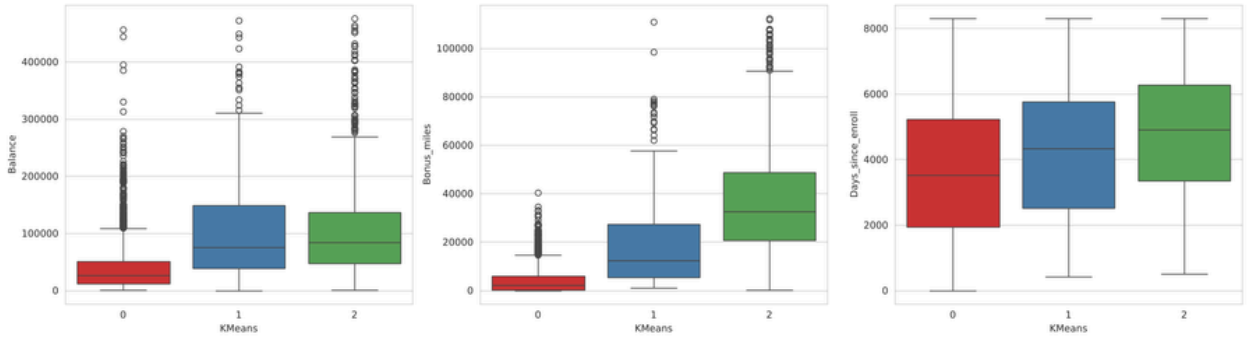


Figure 11. Boxplots for each K-Means cluster (x-axis: cluster, y-axis: Balance, Bonus_miles and Days_since_enroll respectively).

Interpretation

The boxplots show that cluster 0 has the lowest balance and bonus miles, and clusters 1 and 2 have a similar and much higher balance. Cluster 2 has the highest bonus miles and the highest days since enrolling (the oldest members). A lot of outliers are still above the boxes, mainly for balance, which is normal for such skewed columns.

Cluster interpretation from the notebook:

K-Means cluster interpretation (K = 3):

Cluster	Size	What they look like	My name for it
0	2363 (~62%)	Lowest balance (~40k), very few bonus miles (~4k), almost no flights (0.4 per year), credit card bin ~1.2, only ~20% got an award, newest members	**Low activity / casual members**
1	262 (~7%)	Very high flight miles (~2600), ~7.5 flight transactions, highest Qual_miles (~280), high balance (~107k), 67% got an award	**Frequent flyers**
2	1190 (~31%)	High balance (~104k), highest bonus miles (~37k), credit card bin ~3.7, but almost no flying (~184 flight miles), oldest members (~4800 days), 59% got an award	**Credit card / bonus miles earners**

Insights:

- The 3 groups are different in how they earn miles, not only how much. Cluster 1 earns by flying, cluster 2 earns by spending on the credit card and bonus partners.
- Cluster 2 customers are loyal (long time members) but they don't fly much, so the airline could try offers to make them fly more (use their big balance).
- Cluster 1 is small but is the most valuable for the airline's actual flight revenue, so it should be taken care of (Topflight status benefits, upgrades).
- Cluster 0 is more than half of the customers and they are barely active. The airline could try to wake them up with credit card offers or small promos.
- Silhouette is 0.308, which means the clusters are there but overlap a lot, as I guessed from the EDA.

Task 5 - Hierarchical clustering

Task 5.1 - Dendrogram (Ward linkage)

Figure12:DendrogramwithWardlinkage(last30 merges)

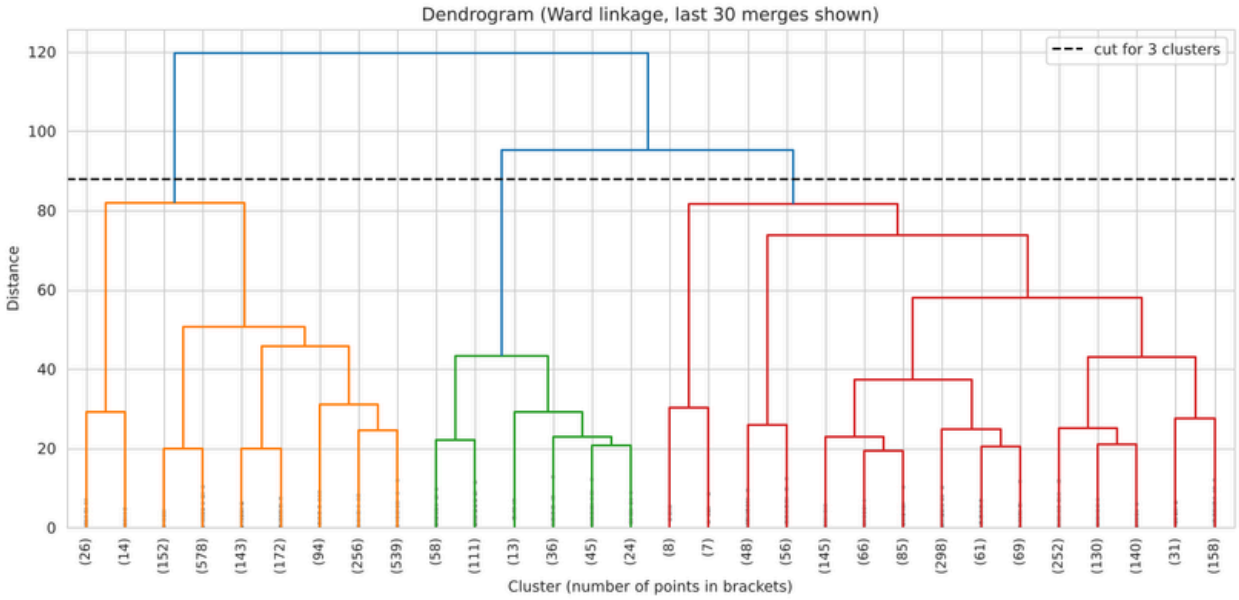


Figure 12. x-axis: clusters, with the number of points in brackets; y-axis: distance at which clusters are merged; the dashed line is the cut for 3 clusters.

Interpretation

Reading the dendrogram: The last merges happen at big distances (around 95 and 120 on the y-axis). If I cut the tree at around 88 (red dashed line) I get 3 branches (orange, green and red), and these branches only join much higher up. So the dendrogram also supports using **3 clusters**, which matches what I got from K-Means.

Task 5.2 - Trying different linkage criteria

Table10:Silhouettescoreandclustersizesforeachlinkage and number of clusters (output of the notebook)

Linkage	Clusters	Silhouette	Smallest cluster size	Largest cluster size
ward	3	0.258	287	1974
ward	4	0.269	40	1934
ward	5	0.273	15	1934
ward	6	0.286	15	1934
complete	3	0.623	8	3767
complete	4	0.621	7	3760
complete	5	0.471	7	3644
complete	6	0.480	7	3525
average	3	0.676	2	3801
average	4	0.619	2	3761
average	5	0.618	2	3761
average	6	0.557	2	3718
single	3	0.669	1	3812
single	4	0.673	1	3798
single	5	0.511	1	3772
single	6	0.519	1	3761

Comparing linkages:

- complete, average and single give very high silhouette scores (0.5 to 0.68), which looks great at first, **but** look at the cluster sizes: the biggest cluster has 3700+ customers out of 3815 and the smallest ones have only 1 to 8 customers. So these methods just pick a few extreme customers and put them in their own tiny clusters, and everyone else is in one giant cluster. The high silhouette is misleading here, and this is not a useful segmentation.
- ward has a lower silhouette (about 0.26 to 0.29) but its clusters are much more balanced, because it tries to minimize the variance inside clusters (the same idea as K-Means).

•So I'm using **Ward linkage with 3 clusters** (also matches the dendrogram).

I then fitted the Ward clustering with 3 clusters. The notebook printed a silhouette score of 0.2583 and these cluster sizes:

Table 11: Hierarchical (Ward, 3 clusters) cluster sizes (output of the notebook)

Cluster	Number of customers
0	1974
1	1554
2	287

Task 5.3 - Hierarchical cluster visualization

Figure13:Hierarchicalclusters(Ward,3clusters)onthe PCA plot

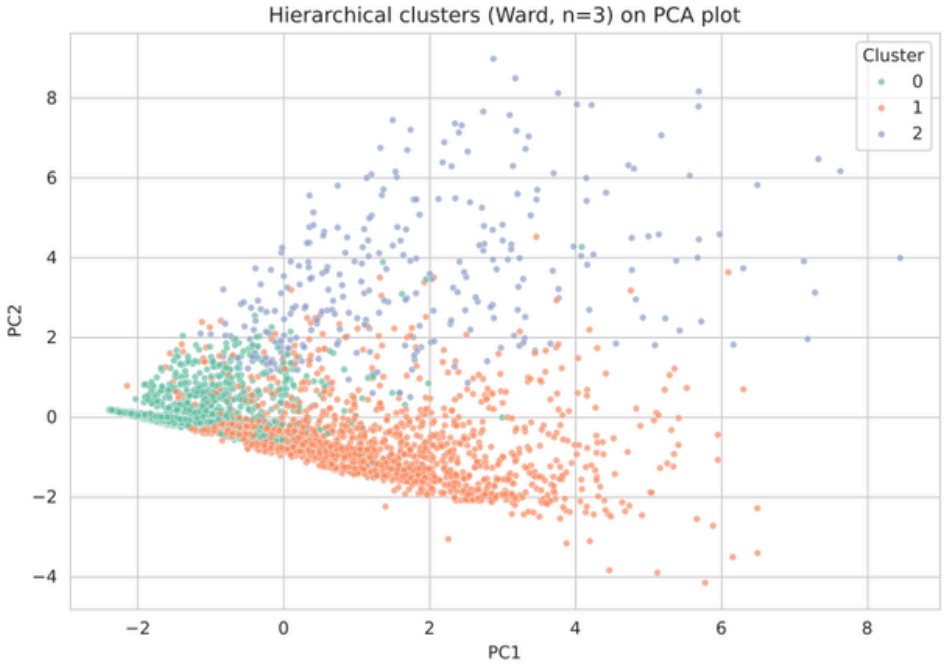


Figure 13. Customers on the first two principal components (x-axis: PC1, y-axis: PC2), coloured by hierarchical cluster.

Interpretation

Here the Ward clusters also overlap. Cluster 0 (green) is at the lower left, cluster 1 (orange) is the large central group going to the right, and cluster 2 (purple) is the sparse group in the upper part. The clusters mix more in the middle than the K-Means ones, which agrees with the lower silhouette score of 0.258.

Task 5.4 - Understanding the hierarchical clusters

Table12:Averagevalueofeachfeatureineachhierarchicalcluster, in the original units (output of the notebook)

Cluster	Balance	Qual_ miles	cc1_ miles	cc2_ miles	cc3_ miles	Bonus_ miles	Bonus_ trans	Flight_ miles_ 12mo	Flight_ trans_ 12	Days_ since_ enroll	Award?	Size
0	40253.8	0.6	1.1	1.0	1.0	3236.2	5.5	84.2	0.3	3800.0	0.2	1974
1	90225.6	119.2	3.3	1.0	1.0	30954.0	17.2	198.0	0.6	4432.6	0.5	1554
2	93889.0	148.8	1.7	1.0	1.0	15702.6	14.8	2370.3	6.9	4125.0	0.7	287

Size = number of customers in the cluster. The averages were rounded to 1 decimal in the notebook.

Figure 14: Hierarchical: average scaled value of each feature per cluster

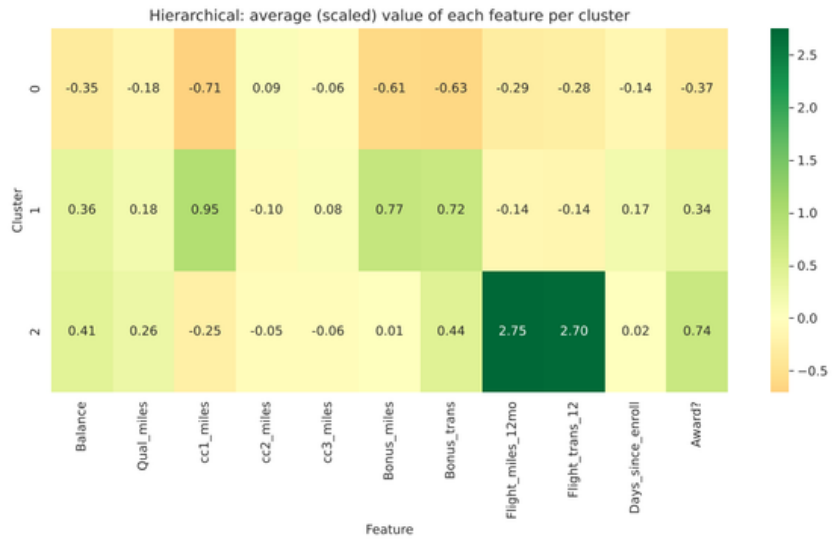


Figure 14. Heatmap of the cluster averages of the scaled features (rows: clusters, columns: features; green = above average, orange = below average).

Interpretation

In this heatmap cluster 2 has very high flight miles and flight transactions (2.75 and 2.70) and the highest award value (0.74). Cluster 1 has the highest credit card bin (0.95), bonus miles (0.77) and bonus transactions (0.72) but low flights (-0.14). Cluster 0 is below average on most columns (for example credit card bin -0.71 and bonus miles -0.61).

Cluster interpretation from the notebook:

Hierarchical (Ward) cluster interpretation:

Cluster	Size	What they look like	Similar to K-Means
0	1974 (~52%)	Balance ~40k, bonus miles ~3k, almost no flights, award rate ~20%	Cluster 0 (low activity)
1	1554 (~41%)	Balance ~90k, bonus miles ~31k, credit card bin ~3.3, hardly any flights	Cluster 2 (card / bonus earners)
2	287 (~7.5%)	Flight miles ~2370, ~7 flight transactions, balance ~94k, award rate ~70%	Cluster 1 (frequent flyers)

Insights:

- Hierarchical clustering found the same 3 types of customers as K-Means (casual, card earners, frequent flyers), which makes me more confident that these segments are real.
- The difference is in the boundaries. Ward put more customers (1554) in the card earner group than K-Means (1190), so the "medium" customers are being split in a different place.
- The silhouette score (0.258) is lower than K-Means (0.308).
- One negative thing is that hierarchical clustering is slow for big data, it was fine here with only ~3800 rows.

Task 6 - DBSCAN

DBSCAN needs two parameters: eps (radius of the neighbourhood) and min_samples (minPts, the minimum points needed in the radius to make a core point). I don't want to just guess so I'll use a k-distance plot to get an idea for eps, and then try a grid of values.

Task 6.1 - k-distance plot to pick eps

Figure 15: k-distance plot (k = 10)

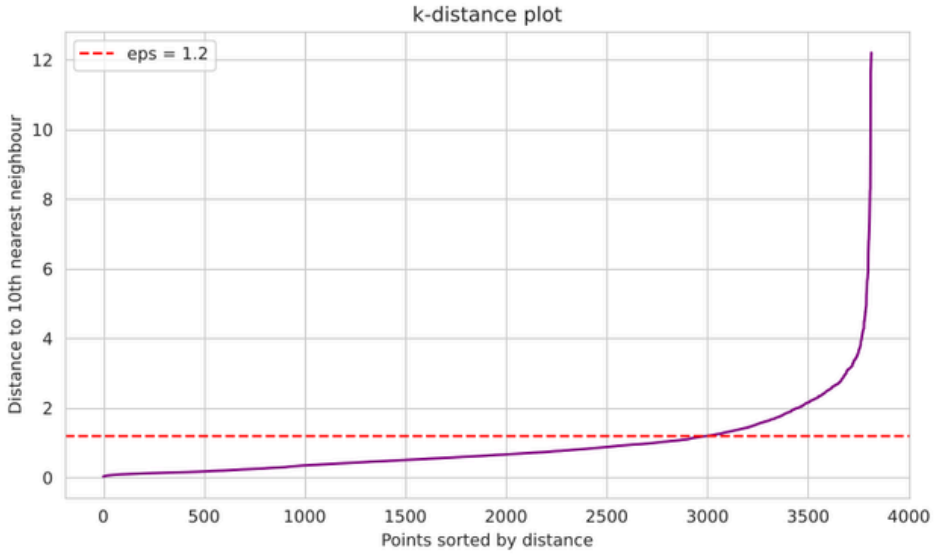


Figure 15. Distance of every customer to its 10th nearest neighbour, sorted from small to large (x-axis: points sorted by distance, y-axis: distance); the red dashed line is eps = 1.2.

Interpretation

Reading the plot: The curve is almost flat up to around 3000 points and after that it starts going up fast. The "knee" is around a distance of 1.2 to 1.5, so I started with eps = 1.2 (red line). Points on the steep right part are far away from everything else, these will become noise. I'll check with the grid below instead of trusting only this plot.

Task 6.2 - Grid search over eps and min_samples

Table13:DBSCANresultsforeachepsandmin_samples(output of the notebook)

eps	min_ samples	clusters	noise points	noise %	silhouette (no noise)
0.6	5 10 15	21 8 9 6	1316 1618	34.5	0.120
0.6	20 5 10	10 5 2 2	1793 1961	42.4	0.185
0.6	15 20 5	5 2 2 2 8	901 1124	47.0	0.178
0.6	10 15 20	3 2 2 6 3	1278 1347	51.4	0.208
0.8	5 10 15	2 2 6 4 3	637 760 874	23.6	0.202
0.8	20 5 10	2	951 474 588	29.5	0.147
0.8	15 20 5		641 674 337	33.5	0.337
0.8	10 15 20		414 474 498	35.3	0.341
1.0			149 228 265	16.7	0.275
1.0			304	19.9	0.302
1.0				22.9	0.317
1.0				24.9	0.323
1.2				12.4	0.256
1.2				15.4	0.292
1.2				16.8	0.294
1.2				17.7 8.8	0.298
1.5				10.9	0.211
1.5				12.4	0.285
1.5				13.1 3.9	0.285
1.5				6.0 6.9	0.286
2.0				8.0	0.258
2.0					0.269
2.0					0.271
2.0					0.271

The silhouette is calculated without the noise points. It is empty (nan) when DBSCAN found only one cluster or none.

Figure 16: Number of clusters and percentage of noise for each eps and min_samples

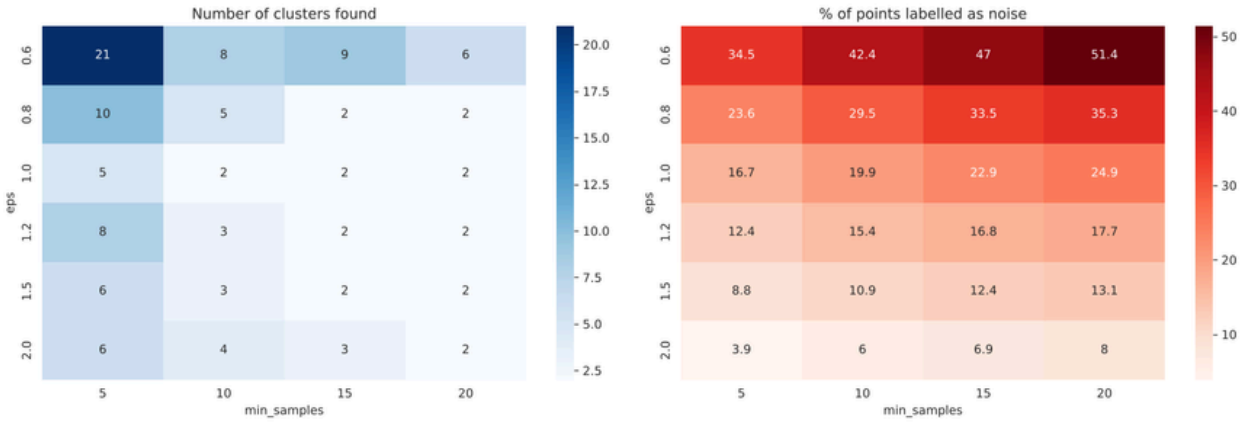


Figure 16. Left: number of clusters found. Right: percentage of points labelled as noise (rows: eps, columns: min_samples).

Interpretation

What happens when I change the parameters:

- With a small eps (0.6, 0.8) there are lots of small clusters and about 24% to 51% of the data is noise. This is too strict.
- With a big eps (2.0) the noise is small (4% to 8%) but the clusters start merging together.
- If I increase min_samples, noise increases and the number of clusters goes down.
- Some settings have a nicer silhouette (0.30 to 0.34) but they only find 2 clusters and throw away 20% to 35% of customers as noise, so that is not really better.
- I'm choosing **eps = 1.2 and min_samples = 10**: 3 clusters, 15.4% noise and a silhouette of 0.29. It's a good middle point.

I chose eps = 1.2 and min_samples = 10. The notebook printed 3 clusters, 588 noise points (15.4%) and a silhouette score of 0.2923 (noise excluded). The cluster sizes were:

Table 14: DBSCAN (eps = 1.2, min_samples = 10) cluster sizes (output of the notebook)

Cluster	Number of customers
-1 (noise)	588
0	2248
1	969
2	10

Task 6.3 - DBSCAN visualization

Figure17:DBSCANclusters(eps=1.2,min_samples = 10) on the PCA plot

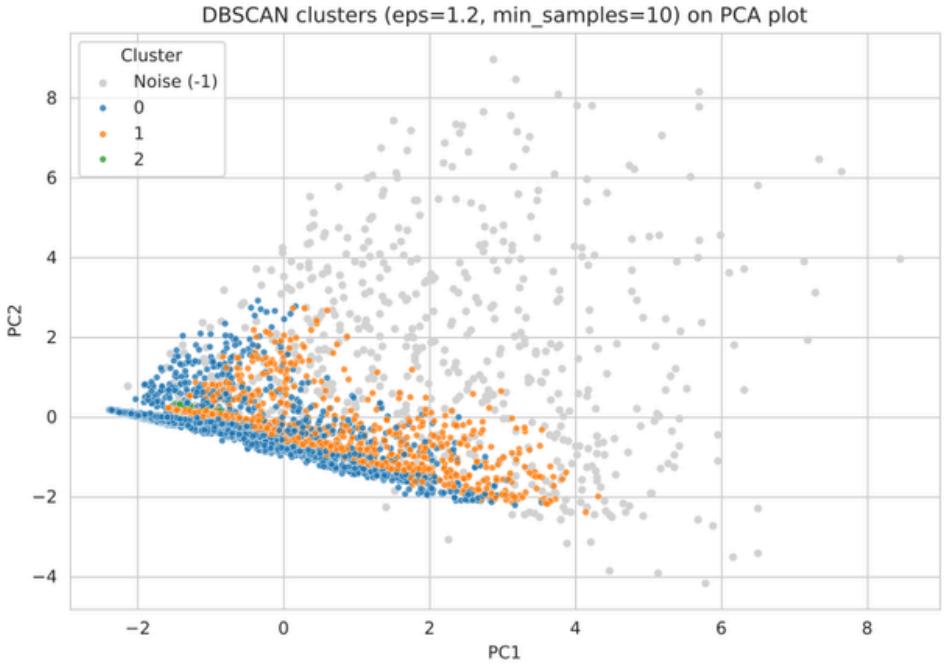


Figure 17. Customers on the first two principal components (x-axis: PC1, y-axis: PC2), coloured by DBSCAN cluster; grey points are noise (-1).

Interpretation

In this 2D view the two big clusters (blue 0 and orange 1) look mixed together in the lower left part, and the grey noise points are spread over the upper and right parts, where the customers are far from each other. Cluster 2 (green) is only a few points at the left side. So the DBSCAN clusters are not easy to see in the PCA plot, and the profile table below explains what separates them.

Task 6.4 - Understanding the DBSCAN clusters

Table15:AveragevalueofeachfeatureineachDBSCANcluster, in the original units (output of the notebook)

Cluster	Balance	Qual_ miles	cc1_ miles	cc2_ miles	cc3_ miles	Bonus_ miles	Bonus_ trans	Flight_ miles_ 12mo	Flight_ trans_ 12	Days_ since_ enroll	Award?	Size
-1	129756.5	389.6	2.5	1.1	1.1	29606.9	18.2	1369.3	4.1	4580.2	0.6	588
0	51895.6	0.0	1.7	1.0	1.0	8918.0	8.3	85.1	0.3	3798.9	0.0	2248
1	55093.6	0.0	2.6	1.0	1.0	22142.5	12.6	162.5	0.6	4449.0	1.0	969
2	27372.7	0.0	1.0	2.0	1.0	8459.1	11.7	0.0	0.0	2919.1	0.0	10

Cluster -1 is the noise group. Size = number of customers. The averages were rounded to 1 decimal in the notebook.

Cluster interpretation from the notebook:

DBSCAN interpretation:

Cluster	Size	What they look like
-1 (noise)	588 (15.4%)	Highest balance (~130k), highest Qual_miles (~390), lots of flights (flight miles ~1370, ~4 trans), 60% awards. Basically the extreme / heavy users
0	2248	**Everyone has Award = 0**, low balance (~52k), low bonus miles (~9k), almost no flights
1	969	**Everyone has Award = 1**, balance ~55k, bonus miles ~22k, higher credit card bin
2	10	Tiny group, the only one using the 2nd credit card (cc2 bin = 2), no flights at all, newest members

Insights:

- DBSCAN mostly split the customers using the Award? column. This is because it's a 0/1 column, so after scaling the distance between award and non-award customers is about 2 in this one column, bigger than eps = 1.2, so the points can't connect across. So the clusters are basically "no award" vs "got an award".
- The very active customers (frequent flyers, high balance) are spread out and not dense, so DBSCAN calls them noise instead of making a cluster. That's actually a weakness for this dataset, because these are probably the customers the airline cares about most. But it is also useful as an outlier detector.
- Cluster 2 (10 customers) is too small to matter much, but it shows there is a small group using the Rewards credit card.
- The silhouette (0.292) is calculated **without** the noise points, so it is not exactly fair to compare with K-Means.

Task 7 - Comparing the three algorithms

Table16:ComparisonofK-Means,HierarchicalandDBSCAN (output of the notebook)

Algorithm	Clusters	Noise points	Silhouette score
K-Means	3	0	0.308
Hierarchical (Ward)	3	0	0.258
DBSCAN	3	588	0.292

Figure 18: The three clusterings side by side on the PCA plot

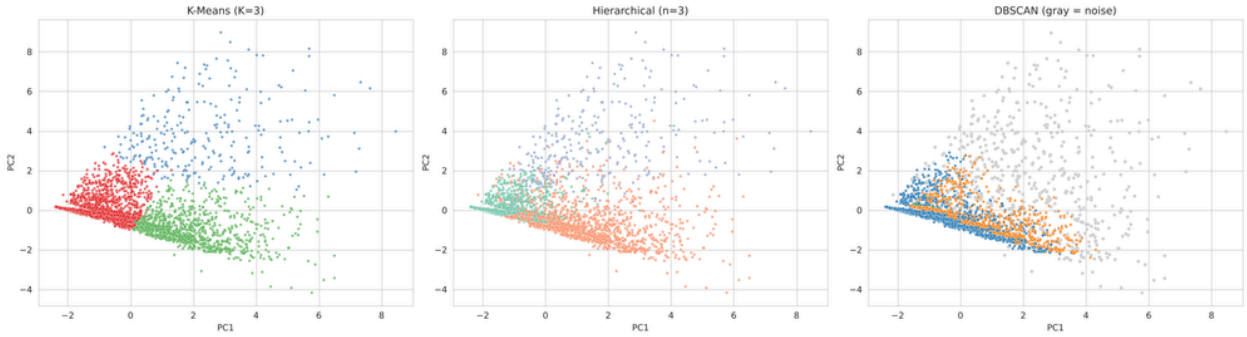


Figure 18. Left: K-Means (K = 3). Middle: Hierarchical (Ward, n = 3). Right: DBSCAN, grey = noise. All use the same axes (x-axis: PC1, y-axis: PC2).

Interpretation

Comparison:

- All three silhouette scores are between 0.26 and 0.31. That is a weak to moderate structure (below 0.5), so the customer groups overlap and the data is more like one continuous cloud than separate islands.
- **K-Means** has the best silhouette (0.308), gives non-overlapping clusters that cover every customer, and the clusters are easy to explain.
- **Hierarchical (Ward)** found almost the same 3 customer types, so it agrees with K-Means. It has the lowest silhouette (0.258), and the dendrogram is useful to see the structure.
- **DBSCAN** found something different (split by award, with the heavy users as noise). Its silhouette (0.292) ignores 15% of the points so it can't be compared directly.
- For a marketing team who wants segments to send offers to, I think **K-Means with K = 3** is the best choice here.

5. Business Recommendations and Conclusion

Task 8 - Final insights and conclusion

Customer segmentsfound (K-Means andHierarchical agree on these):

1. **Low activity members (about 50 to 60% of customers):** small balance, few bonus miles, hardly fly, low award rate.
2. **Credit card / bonus miles earners (about 30 to 40%):** big balance mostly from credit card and bonus miles, old members, but don't fly much.
3. **Frequent flyers (about 7%):** lots of flight miles and flight transactions, highest qualifying miles, high award rate.

What the airline could do:

- Frequent flyers: protect them with premium benefits, upgrades, Topflight status.
- Card earners: they have many miles stored but don't use the airline, so give them offers or promotions to redeem miles and fly more.
- Low activity members: this is the biggest group, so small promotions or credit card sign-up offers could make a big difference.

What I learned about the methods:

- Scaling was really important, and the z-score method worked better than the IQR method for removing outliers on this skewed data.
- The silhouette score alone can fool you (the complete / average / single linkages had high scores but with useless clusters), so I had to look at the cluster sizes too.
- DBSCAN is very sensitive to eps and min_samples, and a binary column can control the whole result.

Limitations / what I would try next:

- The data is very skewed, so I could try a log transform before scaling.
- cc1/cc2/cc3_miles are bins (categories) and Award? is 0/1, so treating them as normal numbers is not perfect. I could try without Award? or use a method like K-Prototypes.
- Silhouette scores are low overall (about 0.3), so the segments are not sharply separated, the labels I gave are just a simple way to describe them.

Dataset Reference

East-West Airlinesdataset (file: EastWestAirlines.xlsx, sheets "Description" and "data"). Source stated in the file: based upon real business data; company names have been changed. (c) 2016 Galit Shmueli and Peter Bruce.